EE 316

Section 3

Pre Lab 08

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Characteristics of BJT and Amplification Behavior

Introduction:

The purpose of this laboratory is to investigate the characteristics and performance of a small-signal bipolar junction transistor (BJT) amplifier. Using LTSpice, the circuit is simulated to analyze its frequency response, output voltage, and gain. Transient analysis is also performed to identify output clipping at higher frequencies and evaluate circuit modifications intended to reduce distortion.

Simulations and Results:

The small signal BJT amplifier in Figure 1 was built and simulated in LTSpice. Figure 2 shows the BJT amplifier in LTSpice. A 500 mV peak-to-peak signal was applied to the circuit over a range of frequencies from 0.1Hz to 1MHz. The resulting Bode plot is shown in Figure 3. A selection of the output voltage and gain for a subset of frequencies is recorded in Table 1.

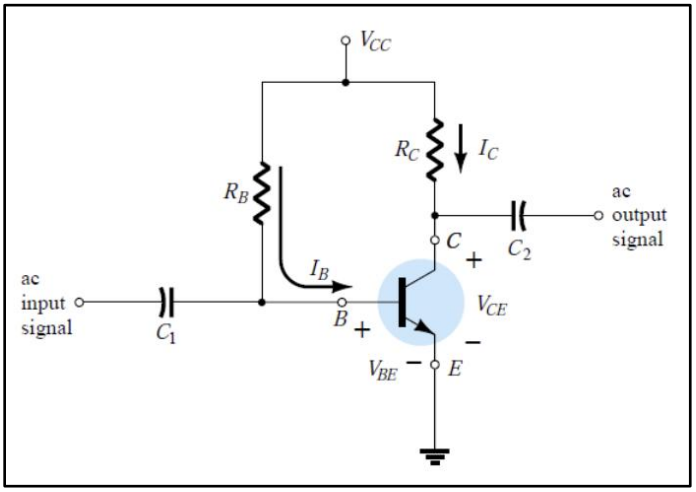


Figure #1: BJT Amplifier (Lab Manual)

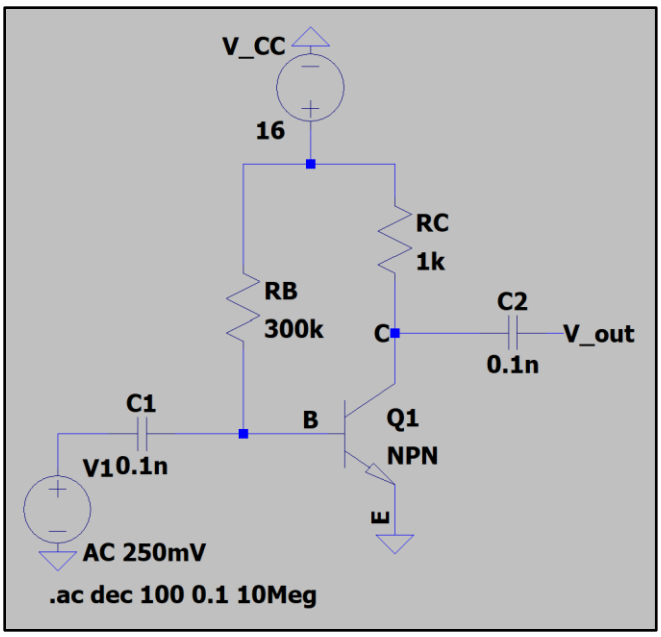


Figure #2: BJT Amplifier Circuit (LTSpice)

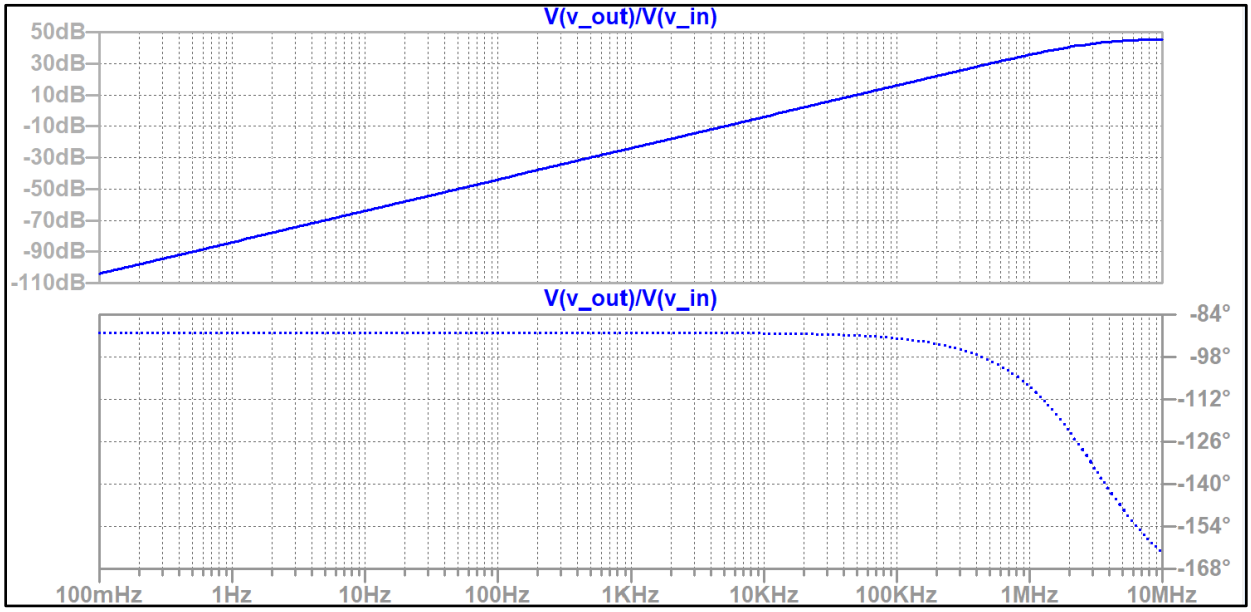


Figure #3: BJT Amplifier Bode Plot

Frequency	V_{out} (V)	Gain (dB)
10 Hz	0.00064	-63.8983
30 Hz	0.00192	-54.3559
60 Hz	0.00383	-48.3353
100 Hz	0.00638	-43.8983
200 Hz	0.01277	-37.8777
500 Hz	0.03192	-29.9189
1 kHz	0.06384	-23.8984

2kHz	0.12767	-17.8781
5 kHz	0.31911	-9.92114
10 kHz	0.63773	-3.90721
15 kHz	0.95538	-0.396496
20 kHz	1.27157	2.08679
50 kHz	3.11309	9.86382
75 kHz	4.53336	13.1284
100 kHz	5.81514	15.2912
150 kHz	7.92018	17.9747
200 kHz	9.46204	19.5197
500 kHz	12.89258	22.2068
750 kHz	13.51964	22.6193
1 MHz	13.76179	22.7735
1.5 MHz	13.94232	22.8867
2 MHz	14.00683	22.9268

Table #1: Simulation Output Voltage and Gain

Figures 4 and 5 show the transient output of the circuit at 10 Hz and 20 MHz respectively. Figure 5 shows significant clipping as the output waveform reaches 16 $V_{pk\text{-}pk}$. After running the simulation at a variety of frequencies, the clipping seems to first occur at around 400 kHz, as shown in Figure 6.

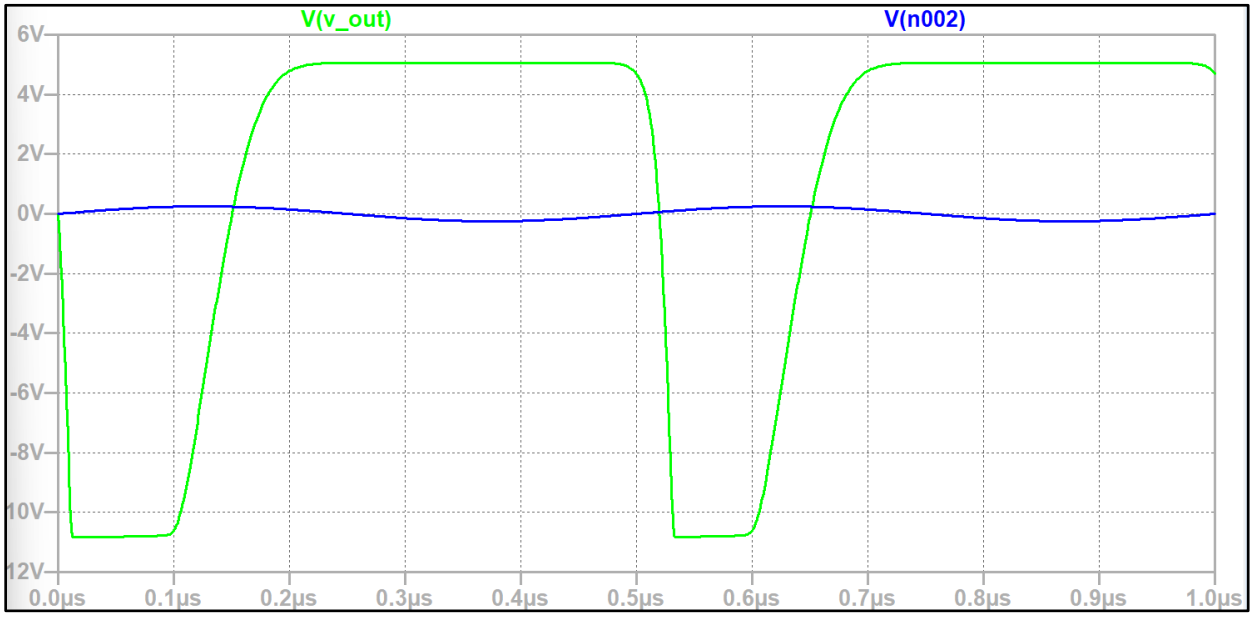


Figure #4: Transient Output at 10 Hz

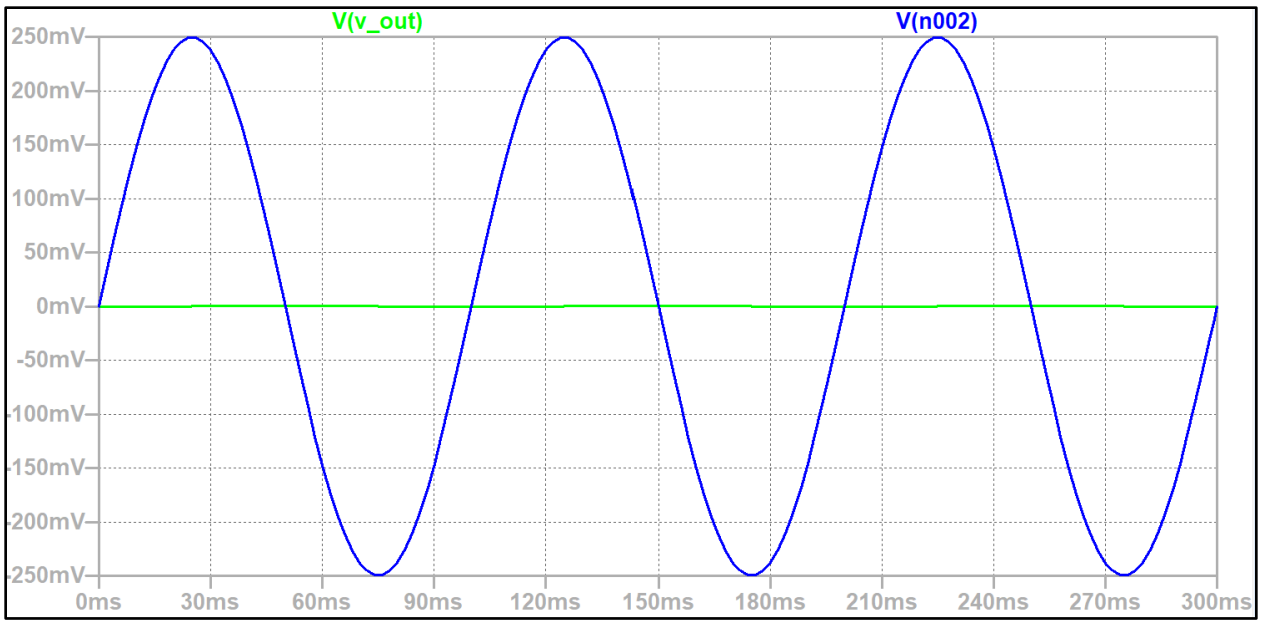


Figure #5: Transient Output at 2 MHz

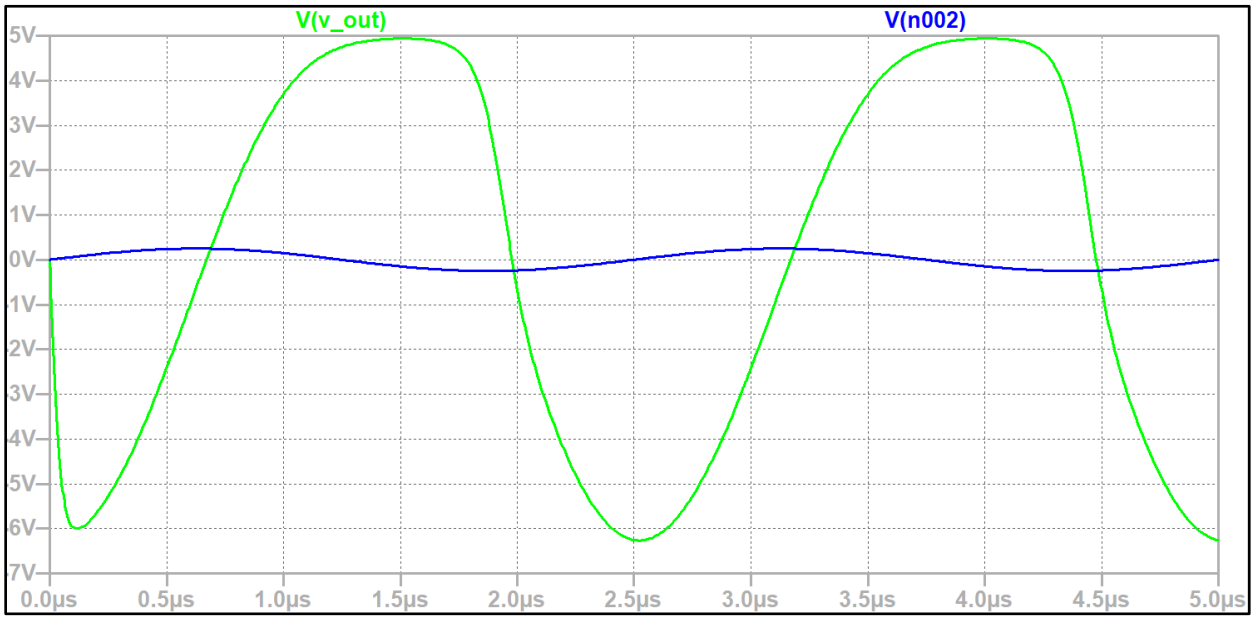


Figure #6: Transient Output at 400 kHz

To reduce this clipping, R_B was halved and C_1 was reduced by 0.01 uF. The modified circuit is shown in Figure 7. The transient output of the modified circuit at 2 MHz is shown in Figure 8.

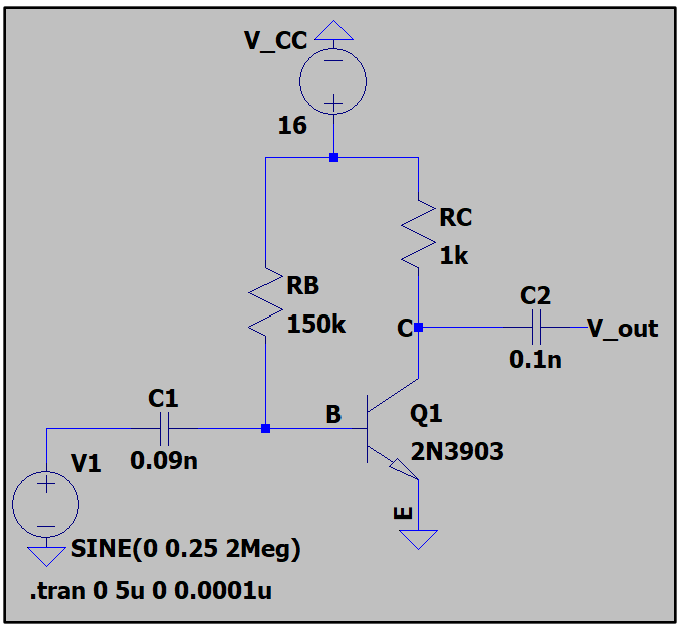


Figure #7: Modified BJT Amplifier Circuit (LTSpice)

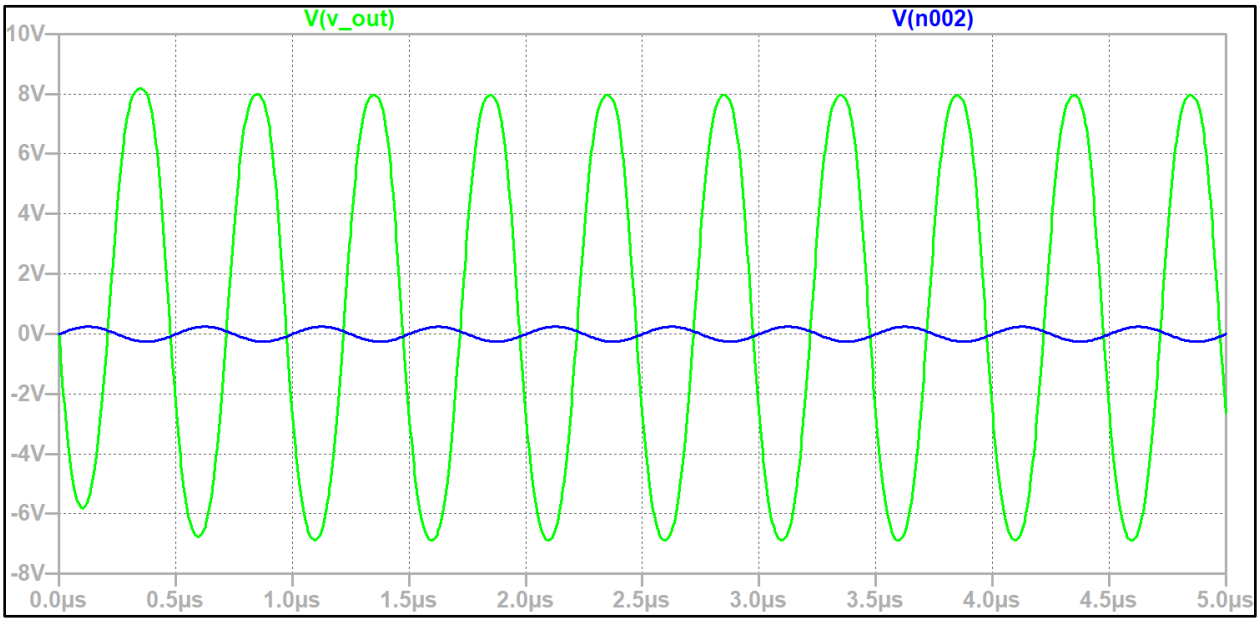


Figure #8: Modified Circuit Transient Output at 2 MHz